Informatica® Cloud Application Integration
April 2025

Azure AI Search Ingestion Automation with ADLS

Informatica Cloud Application Integration Azure AI Search Ingestion Automation with ADLS
April 2025

© Copyright Informatica LLC 2024, 2025

This software and documentation contain proprietary information of Informatica LLC and are provided under a license agreement containing restrictions on use and disclosure and are also protected by copyright law. Reverse engineering of the software is prohibited. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC. This Software may be protected by U.S. and/or international Patents and other Patents Pending.

Use, duplication, or disclosure of the Software by the U.S. Government is subject to the restrictions set forth in the applicable software license agreement and as provided in DFARS 227.7202-1(a) and 227.7702-3(a) (1995), DFARS 252.227-7013©(1)(ii) (OCT 1988), FAR 12.212(a) (1995), FAR 52.227-19, or FAR 52.227-14 (ALT III), as applicable.

The information in this product or documentation is subject to change without notice. If you find any problems in this product or documentation, please report them to us in writing.

Informatica, Informatica Platform, Informatica Data Services, PowerCenter, PowerCenterRT, PowerCenter Connect, PowerCenter Data Analyzer, PowerExchange, PowerMart, Metadata Manager, Informatica Data Quality, Informatica Data Explorer, Informatica B2B Data Transformation, Informatica B2B Data Exchange Informatica On Demand, Informatica Identity Resolution, Informatica Application Information Lifecycle Management, Informatica Complex Event Processing, Ultra Messaging, Informatica Master Data Management, and Live Data Map are trademarks or registered trademarks of Informatica LLC in the United States and in jurisdictions throughout the world. All other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties, including without limitation: Copyright DataDirect Technologies. All rights reserved. Copyright © Sun Microsystems. All rights reserved. Copyright © RSA Security Inc. All Rights Reserved. Copyright © Ordinal Technology Corp. All rights reserved. Copyright © Aandacht c.v. All rights reserved. Copyright Genivia, Inc. All rights reserved. Copyright Isomorphic Software. All rights reserved. Copyright © Meta Integration Technology, Inc. All rights reserved. Copyright © Intalio. All rights reserved. Copyright © Oracle. All rights reserved. Copyright © Adobe Systems Incorporated. All rights reserved. Copyright © DataArt, Inc. All rights reserved. Copyright © ComponentSource. All rights reserved. Copyright © Microsoft Corporation. All rights reserved. Copyright © Rogue Wave Software, Inc. All rights reserved. Copyright © Teradata Corporation. All rights reserved. Copyright © Yahoo! Inc. All rights reserved. Copyright © Glyph & Cog, LLC. All rights reserved. Copyright © Thinkmap, Inc. All rights reserved. Copyright © Clearpace Software Limited. All rights reserved. Copyright © Information Builders, Inc. All rights reserved. Copyright © OSS Nokalva, Inc. All rights reserved. Copyright Edifecs, Inc. All rights reserved. Copyright Cleo Communications, Inc. All rights reserved. Copyright © International Organization for Standardization 1986. All rights reserved. Copyright © ej-technologies GmbH. All rights reserved. Copyright © Jaspersoft Corporation. All rights reserved. Copyright © International Business Machines Corporation. All rights reserved. Copyright © yWorks GmbH. All rights reserved. Copyright © Lucent Technologies. All rights reserved. Copyright © University of Toronto. All rights reserved. Copyright © Daniel Veillard. All rights reserved. Copyright © Unicode, Inc. Copyright IBM Corp. All rights reserved. Copyright © MicroQuill Software Publishing, Inc. All rights reserved. Copyright © PassMark Software Pty Ltd. All rights reserved. Copyright © LogiXML, Inc. All rights reserved. Copyright © 2003-2010 Lorenzi Davide, All rights reserved. Copyright © Red Hat, Inc. All rights reserved. Copyright © The Board of Trustees of the Leland Stanford Junior University. All rights reserved. Copyright © EMC Corporation. All rights reserved. Copyright © Flexera Software. All rights reserved. Copyright © Jinfonet Software. All rights reserved. Copyright © Apple Inc. All rights reserved. Copyright © Telerik Inc. All rights reserved. Copyright © BEA Systems. All rights reserved. Copyright © PDFlib GmbH. All rights reserved. Copyright © Orientation in Objects GmbH. All rights reserved. Copyright © Tanuki Software, Ltd. All rights reserved. Copyright © Ricebridge. All rights reserved. Copyright © Sencha, Inc. All rights reserved. Copyright © Scalable Systems, Inc. All rights reserved. Copyright © jQWidgets. All rights reserved. Copyright © Tableau Software, Inc. All rights reserved. Copyright© MaxMind, Inc. All Rights Reserved. Copyright © TMate Software s.r.o. All rights reserved. Copyright © MapR Technologies Inc. All rights reserved. Copyright © Amazon Corporate LLC. All rights reserved. Copyright © Highsoft. All rights reserved. Copyright © Python Software Foundation. All rights reserved. Copyright © BeOpen.com. All rights reserved. Copyright © CNRI. All rights reserved.

This product includes software developed by the Apache Software Foundation (http://www.apache.org/), and/or other software which is licensed under various versions of the Apache License (the "License"). You may obtain a copy of these Licenses at http://www.apache.org/licenses/. Unless required by applicable law or agreed to in writing, software distributed under these Licenses is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the Licenses for the specific language governing permissions and limitations under the Licenses.

This product includes software which was developed by Mozilla (http://www.mozilla.org/), software copyright The JBoss Group, LLC, all rights reserved; software copyright © 1999-2006 by Bruno Lowagie and Paulo Soares and other software which is licensed under various versions of the GNU Lesser General Public License Agreement, which may be found at http:// www.gnu.org/licenses/lgpl.html. The materials are provided free of charge by Informatica, "as-is", without warranty of any kind, either express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

The product includes ACE(TM) and TAO(TM) software copyrighted by Douglas C. Schmidt and his research group at Washington University, University of California, Irvine, and Vanderbilt University, Copyright (©) 1993-2006, all rights reserved.

This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (copyright The OpenSSL Project. All Rights Reserved) and redistribution of this software is subject to terms available at http://www.openssl.org and http://www.openssl.org/source/license.html.

This product includes Curl software which is Copyright 1996-2013, Daniel Stenberg, <daniel@haxx.se>. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://curl.haxx.se/docs/copyright.html. Permission to use, copy, modify, and distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

The product includes software copyright 2001-2005 (©) MetaStuff, Ltd. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://www.dom4j.org/ license.html.

The product includes software copyright © 2004-2007, The Dojo Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http://dojotoolkit.org/license.

This product includes ICU software which is copyright International Business Machines Corporation and others. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://source.icu-project.org/repos/icu/icu/trunk/license.html.

This product includes software copyright © 1996-2006 Per Bothner. All rights reserved. Your right to use such materials is set forth in the license which may be found at http:// www.gnu.org/software/ kawa/Software-License.html.

This product includes OSSP UUID software which is Copyright © 2002 Ralf S. Engelschall, Copyright © 2002 The OSSP Project Copyright © 2002 Cable & Wireless Deutschland. Permissions and limitations regarding this software are subject to terms available at http://www.opensource.org/licenses/mit-license.php.

This product includes software developed by Boost (http://www.boost.org/) or under the Boost software license. Permissions and limitations regarding this software are subject to terms available at http:/ /www.boost.org/LICENSE_1_0.txt.

This product includes software copyright © 1997-2007 University of Cambridge. Permissions and limitations regarding this software are subject to terms available at http:// www.pcre.org/license.txt.

This product includes software copyright © 2007 The Eclipse Foundation. All Rights Reserved. Permissions and limitations regarding this software are subject to terms available at http:// www.eclipse.org/org/documents/epl-v10.php and at http://www.eclipse.org/org/documents/edl-v10.php.

This product includes software licensed under the terms at http://www.tcl.tk/software/tcltk/license.html, http://www.bosrup.com/web/overlib/?License, http://www.stlport.org/doc/ license.html, http://asm.ow2.org/license.html, http://www.cryptix.org/LICENSE.TXT, http://hsqldb.org/web/hsqlLicense.html, http://httpunit.sourceforge.net/doc/ license.html, http://jung.sourceforge.net/license.txt , http://www.gzip.org/zlib/zlib_license.html, http://www.openldap.org/software/release/license.html, http://www.libssh2.org, http://slf4j.org/license.html, http://www.sente.ch/software/OpenSourceLicense.html, http://fusesource.com/downloads/license-agreements/fuse-message-broker-v-5-3- license-agreement; http://antlr.org/license.html; http://aopalliance.sourceforge.net/; http://www.bouncycastle.org/licence.html; http://www.jgraph.com/jgraphdownload.html; http://www.jcraft.com/jsch/LICENSE.txt; http://jotm.objectweb.org/bsd_license.html; . http://www.w3.org/Consortium/Legal/2002/copyright-software-20021231; http://www.slf4j.org/license.html; http://nanoxml.sourceforge.net/orig/copyright.html; http://www.json.org/license.html; http://forge.ow2.org/projects/javaservice/, http://www.postgresql.org/about/licence.html, http://www.sqlite.org/copyright.html, http://www.tcl.tk/software/tcltk/license.html, http://www.jaxen.org/faq.html, http://www.jdom.org/docs/faq.html, http://www.slf4j.org/license.html; http://www.iodbc.org/dataspace/iodbc/wiki/iODBC/License; http://www.keplerproject.org/md5/license.html; http://www.toedter.com/en/jcalendar/license.html; http://www.edankert.com/bounce/index.html; http://www.net-snmp.org/about/license.html; http://www.openmdx.org/#FAQ; http://www.php.net/license/3_01.txt; http://srp.stanford.edu/license.txt; http://www.schneier.com/blowfish.html; http://www.jmock.org/license.html; http://xsom.java.net; http://benalman.com/about/license/; https://github.com/CreateJS/EaselJS/blob/master/src/easeljs/display/Bitmap.js; http://www.h2database.com/html/license.html#summary; http://jsoncpp.sourceforge.net/LICENSE; http://jdbc.postgresql.org/license.html; http://protobuf.googlecode.com/svn/trunk/src/google/protobuf/descriptor.proto; https://github.com/rantav/hector/blob/master/LICENSE; http://web.mit.edu/Kerberos/krb5-current/doc/mitK5license.html; http://jibx.sourceforge.net/jibx-license.html; https://github.com/lyokato/libgeohash/blob/master/LICENSE; https://github.com/hjiang/jsonxx/blob/master/LICENSE; https://code.google.com/p/lz4/; https://github.com/jedisct1/libsodium/blob/master/LICENSE; http://one-jar.sourceforge.net/index.php?page=documents&file=license; https://github.com/EsotericSoftware/kryo/blob/master/license.txt; http://www.scala-lang.org/license.html; https://github.com/tinkerpop/blueprints/blob/master/LICENSE.txt; http://gee.cs.oswego.edu/dl/classes/EDU/oswego/cs/dl/util/concurrent/intro.html; https://aws.amazon.com/asl/; https://github.com/twbs/bootstrap/blob/master/LICENSE; https://sourceforge.net/p/xmlunit/code/HEAD/tree/trunk/LICENSE.txt; https://github.com/documentcloud/underscore-contrib/blob/master/LICENSE, and https://github.com/apache/hbase/blob/master/LICENSE.txt.

This product includes software licensed under the Academic Free License (http://www.opensource.org/licenses/afl-3.0.php), the Common Development and Distribution License (http://www.opensource.org/licenses/cddl1.php) the Common Public License (http://www.opensource.org/licenses/cpl1.0.php), the Sun Binary Code License Agreement Supplemental License Terms, the BSD License (http:// www.opensource.org/licenses/bsd-license.php), the new BSD License (http://opensource.org/licenses/BSD-3-Clause), the MIT License (http://www.opensource.org/licenses/mit-license.php), the Artistic License (http://www.opensource.org/licenses/artistic-license-1.0) and the Initial Developer's Public License Version 1.0 (http://www.firebirdsql.org/en/initial-developer-s-public-license-version-1-0/).

This product includes software copyright © 2003-2006 Joe Walnes, 2006-2007 XStream Committers. All rights reserved. Permissions and limitations regarding this software are subject to terms available at http://xstream.codehaus.org/license.html. This product includes software developed by the Indiana University Extreme! Lab. For further information please visit http://www.extreme.indiana.edu/.

This product includes software Copyright (c) 2013 Frank Balluffi and Markus Moeller. All rights reserved. Permissions and limitations regarding this software are subject to terms of the MIT license.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

Publication Date: 2025-08-30

Table of Contents

Preface 5

Chapter 1: Introduction to Azure AI Search Ingestion Automation with ADLS recipe 6

Azure AI Search Ingestion Automation with ADLS recipe contents 6

Azure AI Search Ingestion Automation with ADLS recipe assets 7

Chapter 2: Using the Azure AI Search Ingestion Automation with ADLS recipe 8

Step 1. Copy and access the recipe content 8

Step 2. Configure and publish the Azure AI Search connection 9

Step 3: Publish the processes 9

Step 4. Invoke the process 10

Preface

Use *Azure AI Search Ingestion Automation with ADLS* to learn how to create an index, data source, and indexer for the existing files in Azure Data Lake Storage (ADLS), populate the index with the data, and then return the documents. This guide assumes that you have an understanding of the Azure AI Search Connector concepts.

CHAPTER 1

Introduction to Azure AI Search Ingestion Automation with ADLS recipe

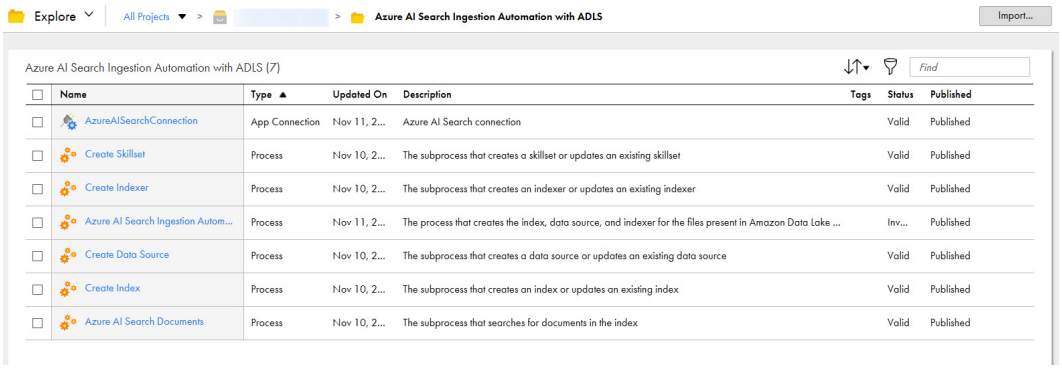
The Azure AI Search Ingestion Automation with ADLS recipe is based on REST and SOAP APIs. Use the recipe to invoke the process and create an index, data source, skillset, and indexer for the existing files in Azure Data Lake Storage (ADLS) container. The process populates the index with the data, and then return the documents that meet the search criteria.

If the index, indexer, and data source input parameters are not provided, the process searches the documents with the index name.

Azure AI Search Ingestion Automation with ADLS recipe contents

The recipe contains multiple assets, such as service connector, app connection, and processes.

The following image shows the assets that the Azure AI Search Ingestion Automation with ADLS recipe package contains:



Azure AI Search Ingestion Automation with ADLS recipe assets

The following table lists the assets that the Azure AI Search Ingestion Automation with ADLS recipe package contains:

Asset Name	Asset Type	Description
AzureAISearchConnection	App connection	Azure AI Search connection.
Create Indexer	Process	Subprocess that creates an indexer or updates an existing indexer.
Create Data Source	Process	Subprocess that creates a data source or updates an existing data source.
Azure AI Search Documents	Process	Subprocess that searches for documents in the index.
Create Index	Process	Subprocess that creates an index or updates an existing index.
Create Skillset	Process	Subprocess that creates a skillset or updates an existing skillset.
Azure AI search ingestion Automation with ADLS	Process	Creates the index, data source, and indexer for the files present in Azure Data Lake Storage (ADLS), populates the index with the data, and then returns the documents that meet the search criteria.

CHAPTER 2

Using the Azure AI Search Ingestion Automation with ADLS recipe

To use the Azure AI Search Ingestion Automation with ADLS recipe, you must perform the following steps manually:

1. Copy and access the recipe
2. Configure and publish the AzureAISearchConnection connection
3. Publish the processes
4. Invoke the process

Step 1. Copy and access the recipe content

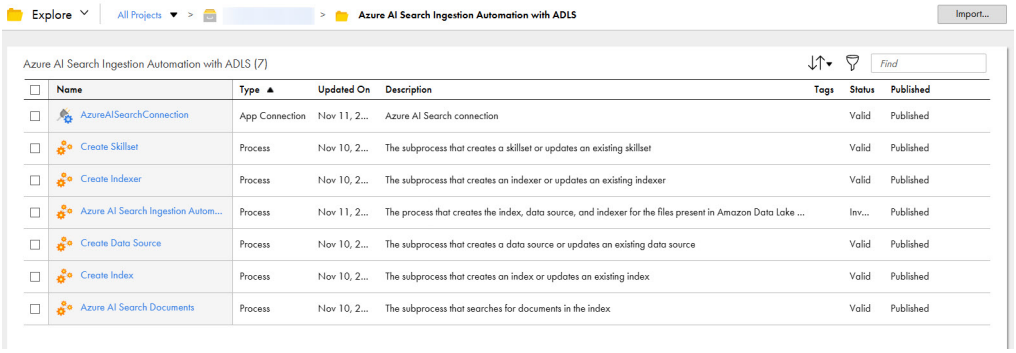
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Azure AI Search Ingestion Automation with ADLS** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Azure AI Search connection

Configure the host name and the API key connection to connect to the service, and then publish the AzureAISearchConnection connection.

1. Open the **AzureAISearchConnection** connection.
2. From the **Type** list, select **Azure AI Search**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Host_Name	Required. The host name of the search service that you want to connect to. You can find the value in **Microsoft Azure Search Service** > **Overview**, and copy the **Url** field value.
API_Key	Required. The admin API key to authenticate Azure AI Search connection requests. You can find the value in **Microsoft Azure search service** > **Settings** > **Keys**, and copy the primary or secondary admin key.

5. Save and publish the connection.

Step 3: Publish the processes

To publish the following processes, click **Actions** in the row that contains the processes in the order specified and select **Publish**.

1. Create Indexer
2. Create Data Source
3. Azure AI Search Documents

4. Create Index
5. Create Skillset
6. Azure AI search ingestion Automation with ADLS

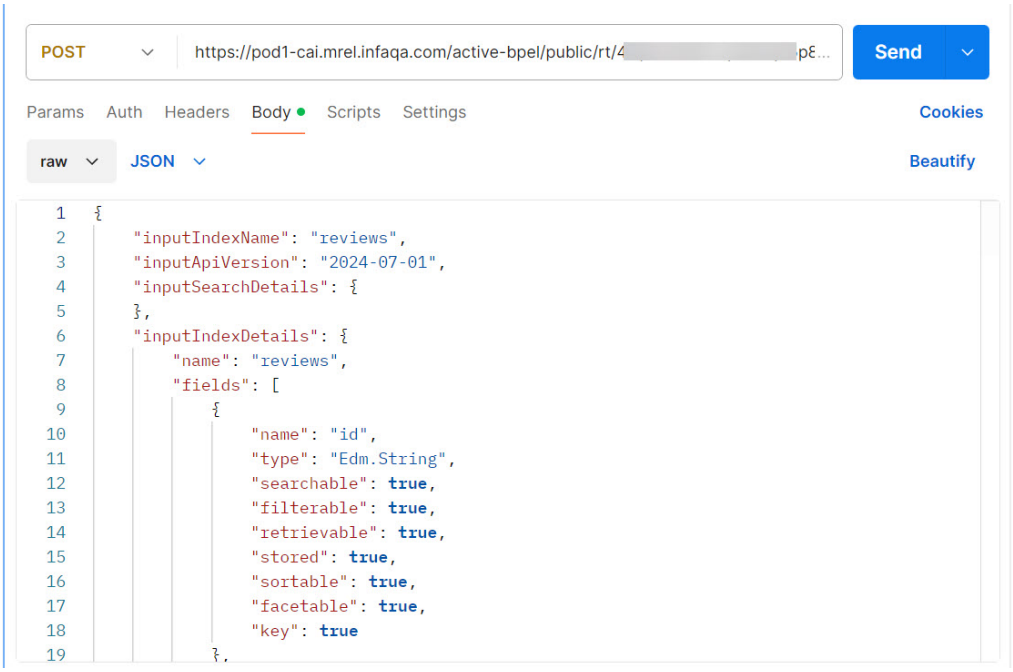
Step 4. Invoke the process

You can invoke the Azure AI Search Ingestion Automation with ADLS process to create an index, data source, skillset, and indexer for the existing files in Azure Data Lake Storage (ADLS), populate the index with the data, and then return the documents that meet the search criteria.

You can pass input through REST or SOAP API endpoints in any API client such as cURL, Postman, SOAP UI, or through any programming language.

For example:

1. Open the **Azure AI search ingestion Automation with ADLS** process and click **Actions** > **Properties Details** > **Copy Service URL**.
2. Open Postman.
3. Select the HTTP verb as POST and specify the generated REST service URL and payload as shown in the following image:



Here is the sample payload:

```
{
    "inputIndexName": "reviews",
    "inputApiVersion": "2024-07-01",
    "inputSearchDetails": {
    },
    "inputIndexDetails": {
        "name": "reviews",
        "fields": [
```

```
        {
            "name": "id",
            "type": "Edm.String",
            "searchable": true,
            "filterable": true,
            "retrievable": true,
            "stored": true,
            "sortable": true,
            "facetable": true,
            "key": true
        },
        {
            "name": "Location",
            "type": "Edm.String",
            "searchable": true,
            "filterable": true,
            "retrievable": true,
            "stored": true,
            "sortable": true,
            "facetable": true
        },
        {
            "name": "Review",
            "type": "Edm.String",
            "searchable": true,
            "filterable": true,
            "retrievable": true,
            "stored": true,
            "sortable": true,
            "facetable": true
        },
        {
            "name": "Date",
            "type": "Edm.String",
            "searchable": true,
            "filterable": true,
            "retrievable": true,
            "stored": true,
            "sortable": true,
            "facetable": true
        },
        {
            "name": "vectorcontent",
            "type": "Collection(Edm.Single)",
            "searchable": true,
            "filterable": false,
            "retrievable": true,
            "stored": true,
            "sortable": false,
            "facetable": false,
            "key": false,
            "dimensions": 1536,
            "vectorSearchProfile": "vector-profile-1727966526693"
        }
    ],
    "similarity": {
        "@odataType": "#Microsoft.Azure.Search.BM25Similarity"
    },
    "semantic": {
        "configurations": [
            {
                "name": "default",
                "prioritizedFields": {
                    "titleField": {
                        "fieldName": "Review"
                    },
                    "prioritizedContentFields": [
                        {
                            "fieldName": "Review"
```

```
                            }
                        ],
                        "prioritizedKeywordsFields": []
                    }
                }
            ]
        },
        "vectorSearch": {
            "algorithms": [
                {
                    "name": "vector-config-1727966528896",
                    "kind": "hnsw",
                    "hnswParameters": {
                        "metric": "cosine",
                        "m": 4,
                        "efConstruction": 400,
                        "efSearch": 500
                    }
                }
            ],
            "profiles": [
                {
                    "name": "vector-profile-1727966526693",
                    "algorithm": "vector-config-1727966528896",
                    "vectorizer": "vectorizer-1727966531637",
                    "compression": "vector-1727966521527-compressor"
                }
            ],
            "vectorizers": [
                {
                    "name": "vectorizer-1727966531637",
                    "kind": "azureOpenAI",
                    "azureOpenAIParameters": {
                        "resourceUri": "https://chatgptopenaiseas.openai.azure.com",
                        "deploymentId": "adavectordb",
                        "apiKey": "5df4e4f520ed4e6f810f0eb070a09680",
                        "modelName": "text-embedding-ada-002"
                    }
                }
            ],
            "compressions": [
                {
                    "name": "vector-1727966521527-compressor",
                    "kind": "scalarQuantization",
                    "rerankWithOriginalVectors": true,
                    "defaultOversampling": 4,
                    "scalarQuantizationParameters": {
                        "quantizedDataType": "int8"
                    }
                }
            ]
        }
    },
    "inputDataSourceDetails": {
        "name": "reviews-datasource",
        "description": "review datasource",
        "type": "adlsgen2",
        "credentials": {
            "connectionString":
"DefaultEndpointsProtocol=https;AccountName=<Search_Index_Host>;AccountKey=<Account_K
ey>;EndpointSuffix=core.windows.net"
        },
        "container": {
            "name": "reviews"
        }
    },
    "inputIndexerDetails": {
        "name": "review-indexer",
        "description": "review indexer",
```

```
            "dataSourceName": "reviews-datasource",
            "targetIndexName": "reviews",
            "parameters": {
                "configuration": {
                    "dataToExtract": "contentAndMetadata",
                    "parsingMode": "default"
                }
            }
        }
    }
```

For more information about the index, indexer, skillset, and data source, see the Azure documentation.

4. Enter the user account details on the **Authorization** tab.
5. Click **Send**.

You can also invoke the process using the Run Using option.